

Chess Engine

Classical negamax chess AI you can play in the browser

Portfolio

Proud

Repo / Path	asadbek/AI ML/4-chess bot
Domain	Game AI / algorithms
Stack	Pure-Python engine (python-chess) + FastAPI; React 18 + Vite + Tailwind
Status	Working - ~90% complete
Live / Deploy	Not publicly deployed

Overview

Play chess in the browser against a custom Python engine. Despite its folder name, this is a strong classical search engine, not a neural network.

Key Features & Technical Highlights

- ~727-line engine: iterative-deepening negamax, alpha-beta, transposition table, null-move pruning, late-move reductions, check extensions, quiescence search.
- Tapered bitboard evaluation; opening book; Boltzmann-sampled difficulty.

Engineering Evidence & Metrics

- Substantial, correct classical-AI implementation - strong CS/algorithms evidence.

Why It Matters (application / interview talking points)

- Great algorithms showcase (search, pruning, evaluation) - frame it as classical game AI, not ML.

Honest Caveats

- Legacy torch/.pth files are unused; README was corrected to reflect the real negamax engine.